

Implementation and Control

CSE 325

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External (Interference) noise

- The largest sources of noise
- Originates outside of the measurement device and is coupled to it.

Sources of interference can be many:



Electrical: Electrical power supply lines, electrical machines, lighting equipment, commutating devices, radio communication transmitters, electrical/atmospheric discharges or cosmic noises etc.

Mechanical: Vibrations, gravitational forces, acceleration etc.

Environmental: Temperature variations, ionization sources, humidity variations, chemical sources etc.

External (Interference) noise

This is the most important of all interferences. In some cases, a noise is easily identifiable.

Example: a common noise in electrical system, especially those that contain long wires, is a 100 Hz noise, This is due to power lines.

It is type of noise is also a good example of a time-periodic noise. When transient or random- is almost impossible to identify and to correct.

Electrical interference noise can enter the measurement system in four ways:

- ☁ **Through capacitive or electrostatic coupling:** Noise is transferred due to **electric fields** between nearby conductors.
- ☁ **Through inductive coupling:** Noise is induced due to **magnetic fields** created by changing currents.
- ☁ **through direct conductive coupling:** Noise enters through **direct electrical connection**
- ☁ **By radiation (Electromagnetic interference):** Noise travels through **air as electromagnetic waves** and affects circuits.

Electromagnetic Interference (EMI)

Electromagnetic Interference (EMI) is unwanted noise or disturbance caused by electromagnetic signals that affect electronic circuits.

⚡ How EMI Occurs

When AC current flows through a wire → it behaves like a **transmitting antenna**.

Nearby wires → act like **receiving antennas**.

If the receiving wire forms a loop → **current gets induced** (due to electromagnetic coupling) . This unwanted induced current is called **EMI**.

Types of EMI

- 1. External EMI:** Comes from outside the system .Examples: Radio transmitters, Mobile networks etc. These are usually the strongest sources.
- 2. Internal EMI:** Generated **within the system itself** . Happens when: **Circuit components interfere with each other. Different parts** of a high-frequency measurement system (or even different parts of a chip) **act as transmitters and receivers**

Electromagnetic Interference (EMI)

Solution:

1. Reduce Wire Length & Loop Area

Keeping wires **short** and minimizing loop size helps reduce EMI. Smaller loops pick up less interference from surrounding electromagnetic fields.

2. Twisting Wires

Twisting two wires together reduces the loop area between them. This helps **cancel induced noise**, making the system less sensitive to interference

3. Use of Coaxial Cables

Coaxial cables have a **shielded outer layer**. This shielding blocks external electromagnetic waves and prevents radiated interference

4. Filters & Software Techniques

Hardware filters (like capacitors and inductors) remove unwanted noise signals . **Software algorithms** can further reduce noise by processing the signal

Other Sources of Noise

When two different metals are joined, they can behave like a thermocouple and generate a small voltage. This effect is called Seebeck effect noise. It produces noise due to temperature differences between the junctions.

In most situations, this noise is very small and not a major issue. Noise problems are often complex and not clearly defined. There is no single fixed way to identify them. Finding the source of noise usually requires careful investigation and testing.

Elimination of Noise

1. Hardware Signal Filtering: Characteristics of different types of filters are shown in a previous class.

•**Filtering** is the **process of removing a certain band of frequencies** from a signal and permitting others to be transmitted.

•**The Pass Band:** the range of frequencies passed by the filter

•**The Stop Band:** the range not passed by the filter

•**CUT OFF frequency:** the boundary between stopping and passing . Technically CUTOFF frequency is defined as the frequency at which the output voltage is **70.7%** of that in the pass band.

Attenuation dB (decibels) =

$$10 \log \left(\frac{P_{out}}{P_{in}} \right) = 20 \log \left(\frac{V_{out}}{V_{in}} \right)$$

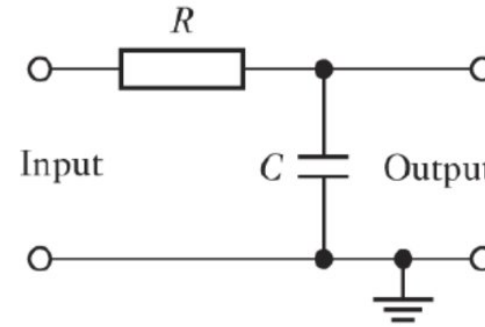


Figure: Passive Low-pass filter

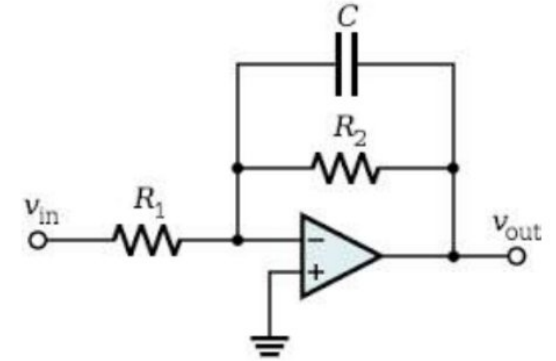


Figure 2: active low-pass filter

Passive Filters

Passive filters are built using only R, L, and C components (resistors, inductors, capacitors). They do **not require external power**. They can **only attenuate (reduce) signals, not amplify**.

Active Filters

Active filters use an operational amplifier (op-amp) along with resistors and capacitors. **They require power supply**. They can **amplify signals as well as filter noise**.

Elimination of Noise

Signal Processing

- Signal processing focuses on improving the quality of the output signal in a measurement system.
- Even after good hardware design, some noise may still remain.

Signal processing helps to reduce or remove this remaining noise

Noise Evaluation

Noise is **random**, so we cannot describe it using a fixed amplitude like a normal signal. Instead, we use **spectral density** $S(f)$, which tells us:

$$S(f) = \frac{V_n^2}{\Delta f}$$

- V_n^2 = noise power
- Δf = bandwidth

So, Spectral density = **noise power per unit bandwidth**



2. Signal-to-Noise Ratio (SNR)

SNR tells us how strong the signal is compared to noise:

$$SNR = \frac{\text{signal power}}{\text{noise power}}$$

Higher SNR = better signal quality

Lower SNR = noisy signal



3. SNR in Decibels (dB)

$$(SNR)_{dB} = 20 \log_{10} \left(\frac{V_s}{V_n} \right)$$

- V_s = signal voltage
- V_n = noise voltage

Data Transmission & Telemetry

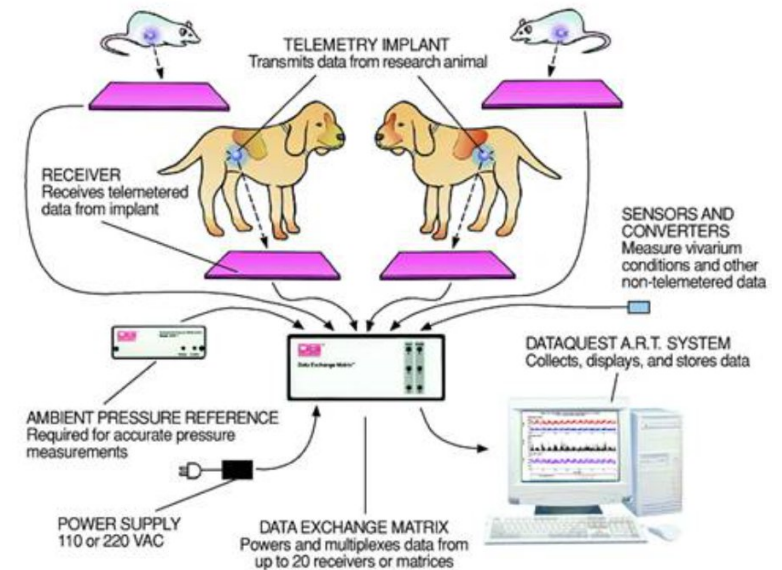
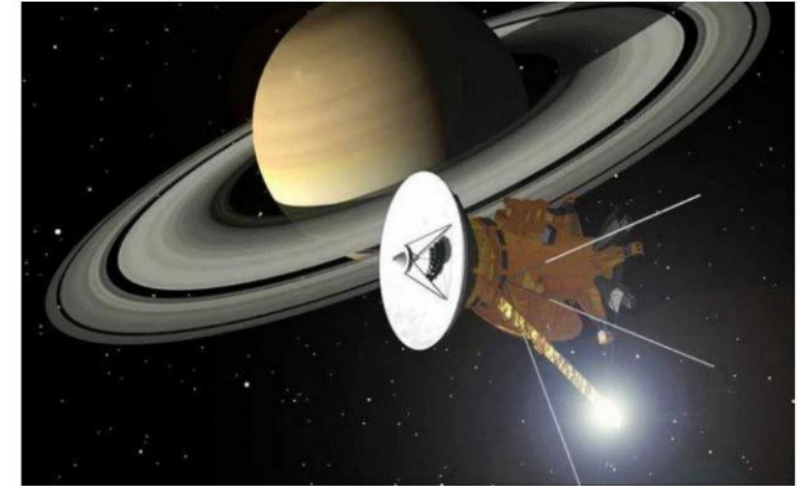
Telemetry – measuring something from distance.

Tele: Greek word; means far away, far off, at a distance)

Example 1: Telemetry from Cassini

Cassini's 12 instruments have returned a daily stream of data from Saturn's system since arriving at Saturn in 2004.

- Telemetry system **does not necessarily mean to employ a long-distance transmission.**
- Can effectively be defined as **measuring something of some place that is not easily accessible or inconveniently located** (e.g. space, animal/human body)



Data Transmission & Telemetry

Formal Definition:

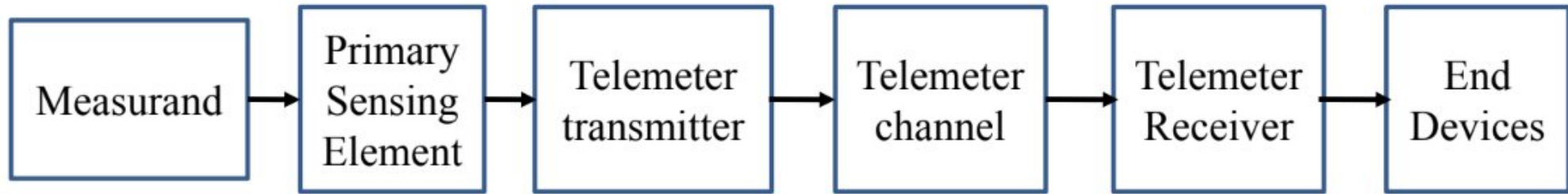
Can be defined as the sensing and measuring of information at some remote/inconvenient location and then transmitting that information to a central or host location. There, it can be monitored and used to control a process at the source site.

If used for control purpose the system is called telecontrol or tele-commandment system

Key aspects:

1. Sensing and measuring a measurand at remote location
2. Transmission to a central or host location
3. The value can be monitored (display, recording, processing, storage) and can be used to control a process at the source location

Functional Block Diagram of a Telemetry System



In the block diagram above, the transmitter and receiver include the signal conditioners.

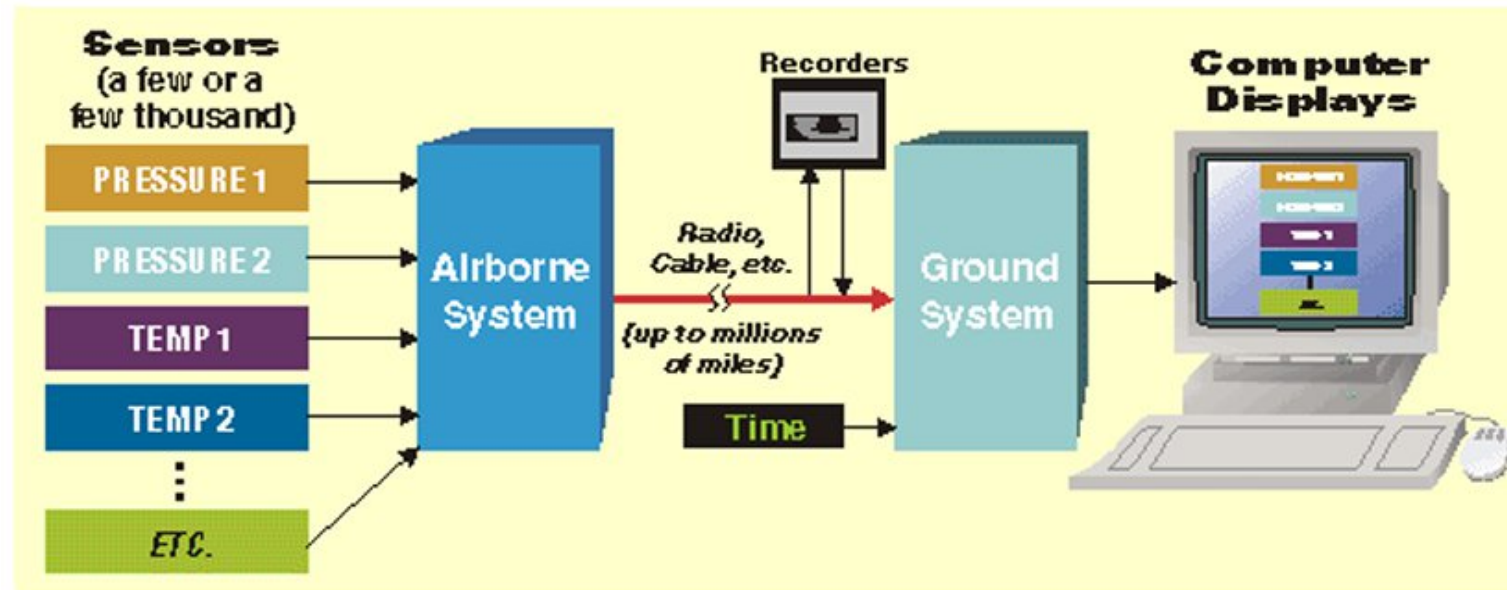


Figure 1: Telemetry system overview.

Functional Block Diagram of a Telemetry System

The three elements peculiar to a telemetry system are:

1. transmitter,
2. channel, and
3. receiver

1. Transmitter

Function: **Make the signal** obtained from the sensor element **suitable for transmission**

2. Channel

Function: **Carry the signal from the transmitter to the receiver station**

3. Receiver

Function: 1. **Receive the signal** sent by the transmitter over the transmission channel



2. **Process the received signal** to make it **suitable for the end device**

Types of Telemetry

Two types of telemetering systems are used based on the channel or media used:

1. wire telemetry (sometimes called land-line telemetry)
2. Wireless telemetry (radio-frequency telemetry)

Methods to be employed depends upon the variable and the distance over which data transmission is needed.

Wire Telemetry:

- Requires **a physical link** between the transmitter and receiver
- May be a cable, existing telephone/telegraph cable or a power line carrier (twisted-pair Cu cable, co-axial cable etc.)

Wireless telemetry:

- **NO physical link/media** is required between the transmitter and receiver
- Link between transmitter and receiver is established through **radio-wave**
- **Carrier modulation** is a must to convey the information to the distant station

Advantages:

- Can **transmit information over longer distances**.
- Can be **used in applications** where measurement points are **at inaccessible or inconvenient locations** (e.g. inside human/animal body and deep space)
- Can **transmit information at higher speeds**

Another Classification of Telemetry

Two types:

1. Base-band telemetry

- **No carrier modulation** is used.
- The **advantage** of base-band telemetry systems is **their simplicity** Because of the base-band transmission, they are normally **limited to only one channel transmission** .
- Can be applied only for wire telemetry

2. Carrier telemetry

- A **high-frequency carrier is modulated** by the measurement signal and transmitted over a wire/wireless media
- The modulated signal **can be transmitted via wired or wireless link**.

Generic Wire-Telemetry System

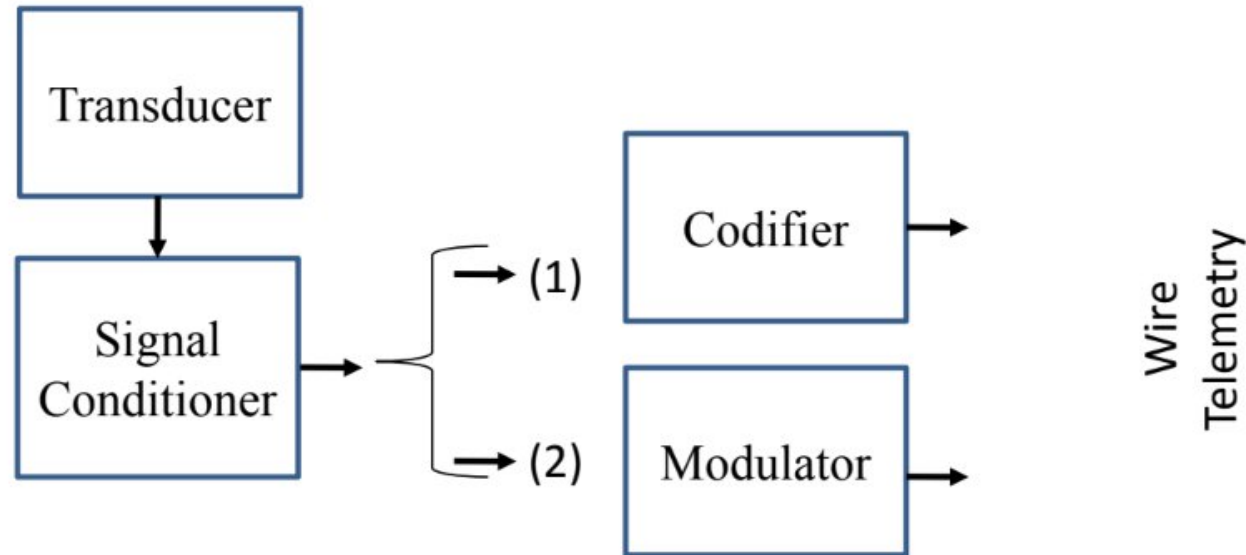


Figure: Generic wire telemetry system; not all the blocks will be always present;
(1) **Base-band telemetry**; (2) **Carrier telemetry**

- Wire telemetry can be performed in either base-band or by sending a modulated signal.

Generic Wire-Telemetry System

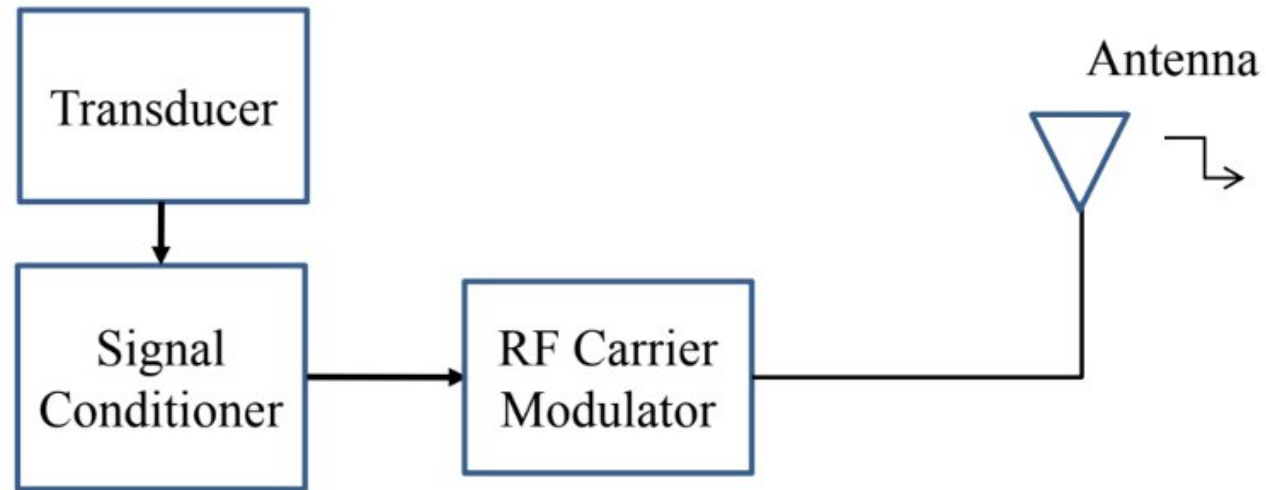
Transmitting End:

1. Transducers: to **convert physical variables** to be measured **into electric signals**
2. Signal Conditioner: Conditioning circuits **to amplify** the low-level signal from the transducer, **limit its bandwidth, and adapt impedance levels;**
3. Codifier: Can be a **digital encoder, V-to-I converter, V-to-f converter** etc.
4. Modulator: A **carrier oscillator whose signal** will be **modulated** by the conditioned signal;

Receiving End (or Base Station): The receiver end consists of similar modules. [\[Draw the receiving end \(also called Base Station\) diagram\]](#)

1. Front-End Conditioner: A generic front end **to amplify the signal** and adapt **the line impedance to the input impedance of the circuits** that follow
2. Decoder: A **digital decoder, I-to-V converter or f-to-V converter** etc.
3. Demodulator: A demodulator circuit using the **demodulation scheme compatible with the transmitter modulation scheme.**
4. End Device

Components of a Typical Wireless Telemetry System



Wireless telemetry always **uses an RF carrier modulation and an antenna.**

RF modulator: An RF carrier is modulated by the conditioned signal;

Antenna: For wireless communication, a transmitting antenna.

Components of a Typical Wireless Telemetry System

Receiving End (or Base Station): The receiver end consists of similar modules.

[Draw the receiving end (also called Base Station) diagram (Homework)]

1. Receiving antenna: Designed for maximum reception at the RF band used
2. Front-End Conditioner: A front-end conditioner to amplify the signal (normally a low noise amplifier, LNA) and to adapt the line impedance to the input impedance of the circuits that follow
3. Radio receiver: A receiver with demodulator circuit using the demodulation scheme compatible with the transmitter modulation scheme.
4. End Device